

Yichen (Eason) Lu

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Education

Carnegie Mellon University – Master of Science in Artificial Intelligence Aug 2023 – Dec 2024

- GPA: 3.76/4.00
- Coursework: Multimodal ML, LLM System, System Tool Chains for AI; Speech Recognition and Understanding

University of Illinois at Urbana-Champaign – BS in Stats & Computer Science Aug 2019 – May 2023

- GPA: 3.93/4.00 (**Highest Honor; Dean's List**)
- Coursework: Computer Vision, Autonomous Vehicle, Applied ML, OS, Linear Algebra, DB Design, Web Dev

Research Interests

Multimodal LLMs, General Speech/Audio LM, Agent-based Translation Systems, Conversational AI

Program Committee: IEEE ASRU 2025, IEEE T-ASLP 2025, AAAI 2026, ICASSP

Work Experience

Research Scientist, Anuttacon – Mountain View, CA Jan 2025 – Present

- Conducted research on **fully duplex speech-to-speech conversational models (Omni-model)**, focusing on continual audio-text pretraining and post-training for comprehensive audio, speech, and music understanding.
- Lead the research and development of general audio understanding models, focusing on Large Audio Language Model (LALM) architectures.
- Constructed and curated a **2M-hour general-purpose speech/audio understanding dataset** with an automated pre-processing pipeline.
- Developed advanced Speech Understanding models for *Whisper from The Star*. [main page]

Publications

[1] **FastAdaSP: Multitask-Adapted Efficient Inference for Large Speech Language Model**

Yichen Lu*, J Song*, CH Yang, Shinji Watanabe
EMNLP 2024 (Oral) [Paper]

[2] **ViDove: A Translation Agent System with Multimodal Context and Memory-Augmented Reasoning**

Yichen Lu*, Wei Dai*, Jiaen Liu*, et al.
EMNLP 2025 System Demonstrations [Paper] [Code]

[3] **SynesLM: A Unified Approach for Audio-visual Speech Recognition and Translation**

Yichen Lu*, J Song*, X Chang, H Bian, S Maiti, Shinji Watanabe
Interspeech 2024 Workshop [Paper]

[4] **Speech-DRAME: A Framework for Human-Aligned Benchmarks in Speech Role-Play**

Jiatong Shi, Jionghao Han, Yichen Lu*, et al.
Under Review

[5] **GALAXY: A Large-Scale Open-Domain Dataset for Multimodal Learning**

Yihan Wu, Yichen Lu, Yijing Chen, Jiaqi Song, William Chen, Ruihua Song, Shinji Watanabe
Interspeech 2025

[6] **Enhancing Audiovisual Speech Recognition by Bifocal Preference Optimization**

Yihan Wu, Yichen Lu, Y Peng, X Wang, R Song, Shinji Watanabe
AAAI 2025 [Paper]

[7] **Robust Audiovisual Speech Recognition Model with Mixture-of-Experts**

Yihan Wu, Yifan Peng, Yichen Lu, X Chang, R Song, Shinji Watanabe
SLT 2024 [Paper]

[8] Exploring Speech Recognition, Translation, and Understanding with Discrete Input

Xuankai Chang, B Yan, K Choi, J Jung, *Yichen Lu*, et al.
ICASSP 2024 [Paper]

[9] Noisy Positive-Unlabeled Learning with Self-Training for Knowledge Graph Completion

Ruijie Wang, B Li, *Yichen Lu*, D Sun, J Li, Yuchen Yan, S Liu, H Tong, T F. Abdelzaher
ACL 2023 Findings [Paper]

Experience

Research Assistant, CMU(WavLab) – Pittsburgh, PA

July 2023 - Jan 2025

- Contribute to open source E2E Speech Processing toolkit ESPNNet (the most popular open-source speech processing repository).
- Research on **Speech Language Models efficient inference** via token merging and pruning.
- Conduct research on a **unified approach for audio-visual language understanding tasks**.
- Conduct research on utilizing SSL discrete token on ASR and ST.

Machine Learning Engineer Intern, TrovaAI – Champaign, IL

May 2023 – Aug 2023

- Designed an **RAG information retrieval pipeline** for the AI agent platform which ingests all types of documents, websites, tables, and live data, and create searchable vector and keyword indices.
- Designed and developed deep link function which will trigger the relevant application based on user's queries for AI agent.
- Designed and developed semantic search and AI assistant platform using **LangChain** and Pinecone Vector Database.
- Designed and developed the evaluation methods for whole AI Agent system via LLM, BLEU, COMET, etc.

Machine Learning Engineer Intern, VMware, Inc. – Beijing, China

July 2021 – Jan 2022

- Individually designed and build up an automatic data analysis system for large-scale machine translation system(BERT-based).
- The system will automatically analyze the machine translation quality and its error rate and generate report monthly.
- Optimized data querying process speed by applying Elastic Stack to large dataset (2 terabyte) management.

Projects

ViDove, *Founder & Researcher*

[Github Repo]

- Developed an End-to-end Multimodal Translation Agent for video subtitle generation in specific field.
- Managed open-source repository of ViDove and directed a technical team with 10 members and a marketing team with about 3 members.
- **IMPACT: ViDove has already officially cooperated with StarCraft 2 World Team League (one of the biggest SC2 E-Sports tournament in the world) for helping them translate the tournament content.**

SUGAR MASSES CREATIVE, *Founder, Developer, & Director*

Feb 2014 - Dec 2018

- Launched **the largest official Minecraft tournament in China (China Minecraft Construction Summit)**, attracting over **1,000 contestants and over 1 million online viewers annually** by leading a team of 30 members.
- Achieved long-term cooperation on Minecraft competitions with major internet companies in China including NetEase, Qihoo360, Tencent, and Youku; secured project investment from NetEase and JoyMe.com.
- Presented at the largest digital entertainment expo in China (ChinaJoy2016, with more than 325,000 entries by business partners, media, consumers, and exhibitors in the four-day event).

Technologies

Languages: Python(Advanced), Java, C/C++ , Rust, R, JavaScript, OCaml, Dart, SQL, CSS, TypeScript

Technologies: PyTorch, Git, LaTeX, PySpark, React.js, Node.js, Flask, PostgreSQL, MongoDB, Linux, Slurm